Problem Set 2: Volatility

MGT 927

Data. CRSP daily data from WRDS, 2 Jan 1990 – 31 Dec 2020 ($T = 7,812$ trading days). The market series is the CRSP value-weighted index return (`vwretd`). The Fama–French `mktrf` series has a 0.998 correlation with it and gives identical conclusions. MSFT is its CRSP holding-period return. All returns are in percent per day. Every likelihood below is coded by hand and maximized numerically (SLSQP, with the stationarity restriction imposed as a constraint). Standard errors are the robust Bollerslev–Wooldridge sandwich $\hat{H}^{-1}\hat{S}\hat{H}^{-1}$, built from a numerical Hessian and the outer product of numerical scores. As an independent check, the `arch` Python package reproduces every point estimate to the third decimal. Code: `volatility_ps.py`; all numbers: `results.json`.

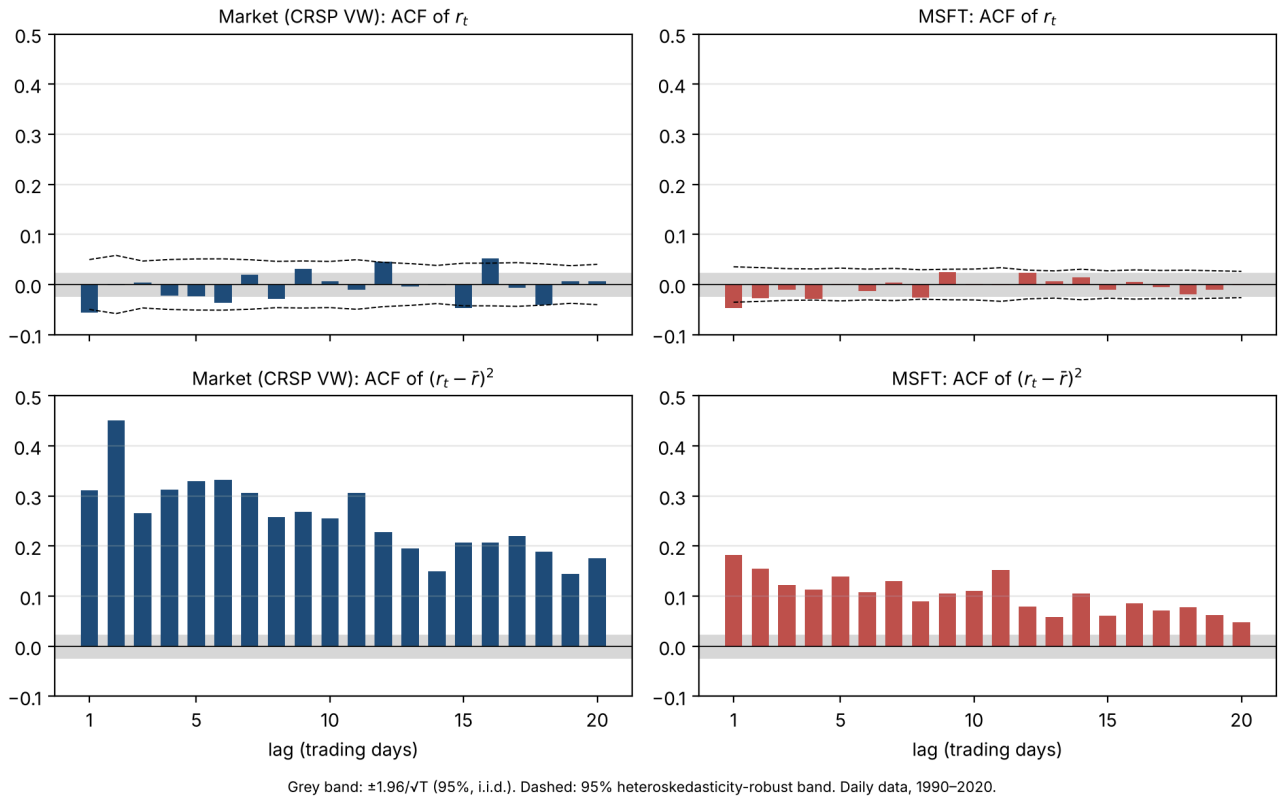
	Mean	S.d.	Ann. vol	Skew	Excess kurt.	Min	Max
Market	0.044	1.13	17.9%	−0.32	11.0	−11.8	11.5
MSFT	0.102	2.02	32.1%	0.23	6.5	−15.6	19.6

Question 1

(a) Autocorrelations of returns and squared returns

Prior (stated before plotting). I expect the *market* to show stronger volatility clustering, meaning higher and more slowly decaying autocorrelations of squared returns. The reasoning starts from the market model, $r_{i,t} = \beta r_{m,t} + e_{i,t}$. The persistent part of volatility is economy-wide: recessions, financial crises, monetary-policy uncertainty and swings in risk appetite keep volatility high for weeks or months, and the market carries all of it. MSFT carries the same market component plus a firm-specific component $e_{i,t}$, which is dominated by one-off news such as earnings releases, product launches and antitrust rulings. That news arrives as a one-day jump and is mostly not followed by more of the same.

Such jumps hurt the measured autocorrelation of squared returns in two ways. They add variance to $r^2_{i,t}$ that does not persist, which dilutes the persistent signal. And an isolated spike enters the denominator of the sample autocorrelation as a fourth power, but the numerator only multiplied by ordinary-sized neighbours, which drags the estimate down. Diversification removes exactly this noise from the index. For the returns themselves I expect autocorrelations close to zero for both series, because predictable returns would be traded away, with perhaps a small negative first-order coefficient from microstructure effects.

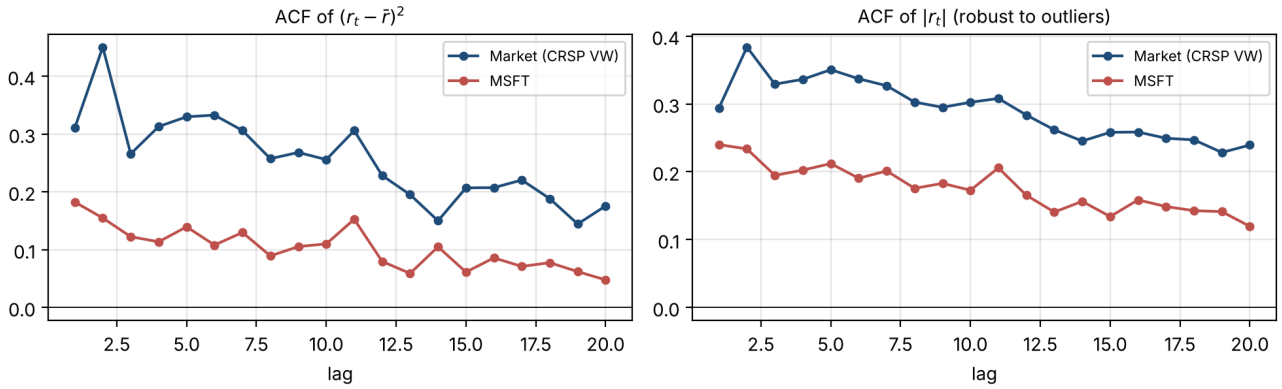


Autocorrelations at lags 1–20. The grey band is the i.i.d. 95% band ±1.96/√T; the dashed lines are the heteroskedasticity-robust 95% band.

Assessment: the prior is correct. Both series show volatility clustering. The autocorrelations of squared (demeaned) returns are positive and outside the 95% band at all 20 lags for both, and the Ljung–Box statistics, Q(20) = 11,056 for the market and 1,853 for MSFT, reject independence overwhelmingly. The clustering is much stronger for the market, on two measures. On level, the market's squared-return autocorrelations average 0.256 over lags 1–20, against 0.103 for MSFT, and the market's Q(20) is six times larger. On persistence, the market's autocorrelation at lag 20 is still 57% of its lag-1 value (0.176 vs 0.311), while MSFT's has fallen to 26% (0.048 vs 0.182).

The ranking is not driven by a handful of crash days. The autocorrelations of absolute returns, which are far less sensitive to outliers, show the same pattern (market 0.29 at lag 1 and 0.24 at lag 20; MSFT 0.24 and 0.12). The data also show the jump mechanism behind the prior directly. MSFT's largest moves are firm-specific: +19.6% on 19 Oct 2000, −14.5% on 3 Apr 2000 and −15.6% on 24 Apr 2000, while the market moved by less than 4% on each of those days. Each of these days is a large squared return that is not followed by persistently large squared returns.

The returns themselves are close to unpredictable, as expected. The largest autocorrelation is −0.056 (market, lag 1). Against the naive i.i.d. band, 10 of the market's 20 lags and 6 of MSFT's look significant. That band is too narrow here, though, because the variance is itself time-varying. Once the band is made robust to heteroskedasticity, only 4 of 20 market lags and 1 of 20 MSFT lags fall outside it, and every coefficient is below 0.06 in absolute value. The economically important dependence is in the second moment, not the first.



Squared-return and absolute-return autocorrelations, market vs MSFT

(b) Gaussian GARCH(1,1) by maximum likelihood

Model.

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \quad z_t \overset{iid}{\sim} N(0,1), \qquad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2,$$

with $\omega > 0$, $\alpha, \beta \geq 0$ and $\alpha + \beta < 1$ (covariance stationarity).

Log likelihood. Conditional on $\mathcal{F}_{t-1}$, $r_t \sim N(\mu, \sigma_t^2)$, and σ_t^2 is a known function of the parameters and past data. By the prediction-error decomposition, the log likelihood is the sum of the one-step conditional densities. With $\theta = (\mu, \omega, \alpha, \beta)$:

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} \ell_t(\theta) = -\frac{1}{2} \sum_{t=1}^{T} \left[\log(2\pi) + \log \sigma_t^2(\theta) + \frac{(r_t - \mu)^2}{\sigma_t^2(\theta)} \right], \qquad \hat{\theta} = \arg\max_{\theta} \mathcal{L}(\theta),$$

with the recursion started at $\sigma_1^2 = T^{-1} \sum_t (r_t - \mu)^2$ (backcast).

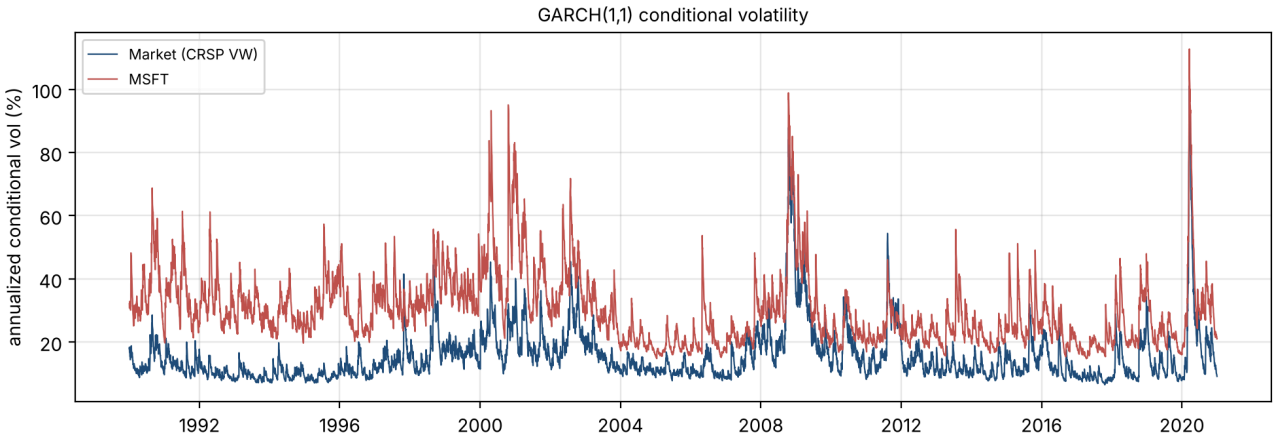
	μ	ω	α	β	$\alpha + \beta$	**Half-life (days)**	**Long-run ann. vol**	**log L**
Market	0.071	0.0175	0.112	0.873	0.985	47	17.4%	−10,110.2
	(0.009)	(0.0035)	(0.012)	(0.012)	(0.005)			
MSFT	0.116	0.066	0.079	0.906	0.985	47	33.9%	−15,649.6
	(0.018)	(0.036)	(0.025)	(0.030)	(0.009)			

Robust standard errors in parentheses. The s.e. of $\alpha + \beta$ uses the delta method with the full robust covariance matrix, including $\mathrm{Cov}(\hat{\alpha}, \hat{\beta})$. *Half-life* $= \log 0.5 / \log(\alpha + \beta)$, *the number of days for a deviation of the conditional variance from its long-run level to halve. Long-run volatility* $= \sqrt{252\,\omega/(1 - \alpha - \beta)}$.

Both series have the classic daily-GARCH shape: a small reaction coefficient α, a large memory coefficient β, and persistence $\alpha + \beta = 0.985$ that is close to one. Whether it is *significantly* below one requires the standard error of the sum, which depends on the covariance of $\hat{\alpha}$ and $\hat{\beta}$ (strongly negative, because a higher α can be offset by a lower β), not just on the two individual standard errors. The estimated correlations are −0.93 for the market and −0.97 for MSFT; ignoring them would inflate the standard error of the sum to 0.017 and 0.039. With that covariance included, the robust standard error of $\alpha + \beta$ is 0.0046 for the market and 0.0086 for MSFT. A one-sided Wald test of the integrated (IGARCH) null $\alpha + \beta = 1$ gives $t = (1 - 0.9854)/0.0046 = 3.2$ for the market ($p < 0.001$), so the market's volatility is clearly mean-reverting. For MSFT, $t = (1 - 0.9855)/0.0086 = 1.7$ ($p \approx 0.045$ one-sided, 0.09 two-sided),

so its persistence is only marginally distinguishable from one, and its long-run variance is correspondingly imprecise. At the point estimates, half of any deviation of the conditional *variance* from its long-run level has dissipated after about 47 trading days, a little over two months, for both series. Where the two differ is in how they get there. The market reacts more strongly to a given return surprise (α = 0.112 vs 0.079), which is the GARCH counterpart of its higher squared-return autocorrelations in part (a): the market's surprises are more informative about its future volatility. MSFT puts more weight on the slowly moving component (β = 0.906). The implied long-run volatilities, 17.4% and 33.9%, are close to the sample volatilities of 17.9% and 32.1%, so the model's unconditional level is consistent with the data.

The model also does what it is supposed to do. On the standardized residuals $\hat{z}_t = \hat{\varepsilon}_t/\hat{\sigma}_t$, the Ljung–Box Q(20) of $\hat{z}_t^2$ falls to 25.2 (p = 0.19) for the market and 11.6 (p = 0.93) for MSFT, from 11,056 and 1,853 for the raw squared returns, so GARCH(1,1) absorbs essentially all of the volatility clustering. What it does not absorb is the fat tails. The standardized residuals have kurtosis of 4.9 (market) and 8.8 (MSFT), against 3 under normality. The Gaussian likelihood is therefore a quasi-likelihood. Its estimates remain consistent as long as the mean and variance equations are right, but the classical standard errors are wrong. That is why the table reports sandwich standard errors; for MSFT they are two to three times the classical ones (for ω: 0.036 vs 0.012).



GARCH(1,1) conditional volatility

(c) "MSFT's volatility is essentially just the market moving"

If the risk manager were right, MSFT's variance would be almost entirely market variance scaled by its beta, $\sigma^2_{MSFT,t} \approx b^2 \sigma^2_{m,t}$. Two things would follow. The market would explain nearly all of MSFT's return variance, and what is left over, the firm-specific residual, would be small, with no volatility dynamics of its own. The estimates contradict both.

Start with how much of MSFT's variance the market explains. In the market model $r_{MSFT,t} = a + b\, r_{m,t} + u_t$, the estimates are $\hat{b}$ = 1.15 (White s.e. 0.02) and R^2 = 0.41, so the market accounts for 41% of MSFT's daily variance over 1990–2020 and firm-specific news for the other 59%. This unconditional split is exact for the sample, and it is the cleanest evidence against the claim.

The GARCH estimates let us look at the same question period by period, but only approximately. Fitting GARCH(1,1) separately to the market and to the residual u_t gives a model-implied variance $b^2\hat{\sigma}^2_{m,t} + \hat{\sigma}^2_{u,t}$ and an **approximate market-related variance share** $b^2\hat{\sigma}^2_{m,t}/(b^2\hat{\sigma}^2_{m,t} + \hat{\sigma}^2_{u,t})$. This is not an exact conditional decomposition of MSFT's variance. That would require a constant beta and a market return that is *conditionally*

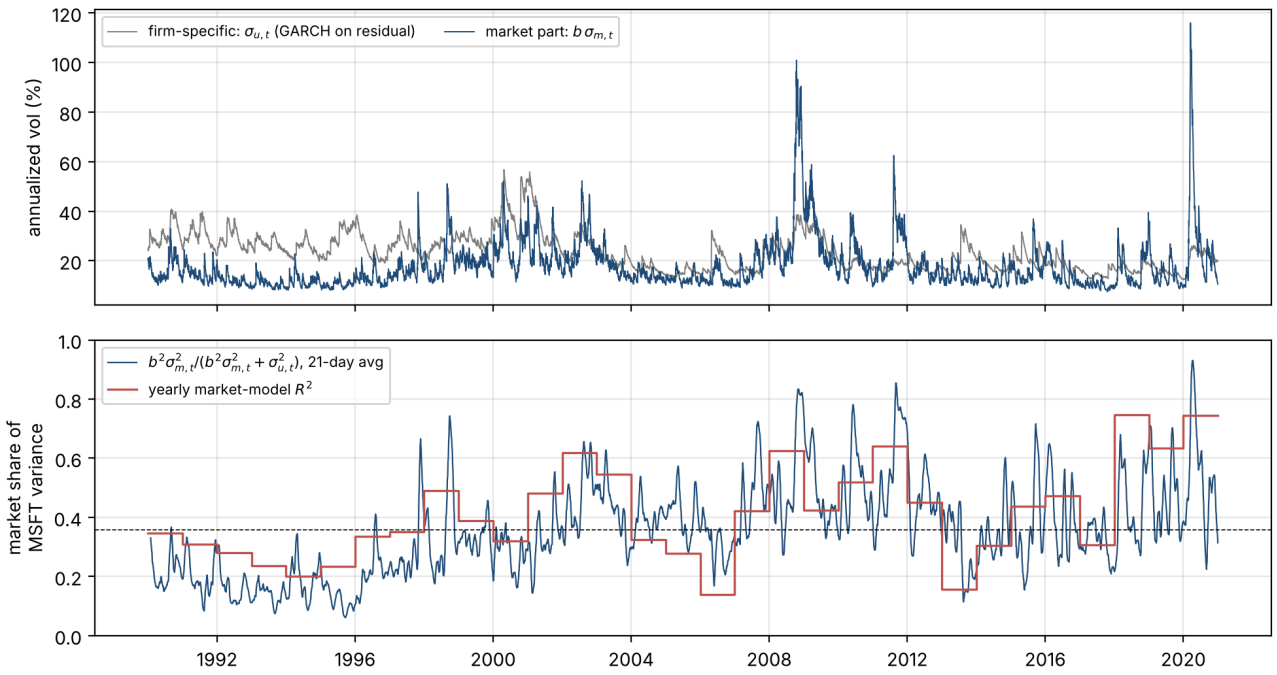
uncorrelated with the residual, whereas OLS only makes them uncorrelated on average, and the yearly betas reported below range from 0.73 to 2.30. As a check on the approximation, the implied total averages 1.05 times MSFT's own GARCH variance, with a 10th–90th percentile range of 0.72–1.43 and a correlation of 0.91 in logs. It therefore tracks MSFT's variance well on average but not exactly on any given day. With that caveat, the approximate share averages only **0.36** and exceeds one half on just 20% of days. In volatility terms, the market-related part averages 17.8% a year and the firm-specific part 23.5%, so the firm-specific part is the larger of the two. A cruder ratio, $\hat{b}_t^2 \hat{\sigma}^2_{m,t} / \hat{\sigma}^2_{MSFT,t}$ with a rolling one-year beta, divides one model's variance by another's. It can exceed one (it does on 2% of days), but its average of 0.41 points the same way.

Next, the residual is far from being noise. GARCH(1,1) on u_t gives α = 0.024 (0.006) and β = 0.971 (0.007), with persistence 0.995 (s.e. 0.002) and a variance half-life of about 150 days, and the Ljung–Box Q(20) on u_t^2 is 704 ($p < 0.001$). Firm-specific volatility therefore clusters on its own, and its estimated persistence is at least as high as the market's. A risk model that simply scaled market volatility by beta would miss this component entirely. The fitted volatility paths make the same point: MSFT's and the market's GARCH volatilities have a correlation of 0.69, so they are related, but less than half of the variation in one is shared with the other.

Period	Approx. market share	Market-related vol	Firm-specific vol
Full sample 1990–2020	0.36	17.8%	23.5%
Dot-com and antitrust 2000–02	0.36	25.3%	33.8%
Financial crisis Sep 2008–Jun 2009	0.64	49.4%	33.7%
COVID crash Mar–Apr 2020	0.87	71.3%	24.2%
Calm year 2017	0.32	10.0%	14.7%

The claim does hold at particular times, and the table shows when. In market-wide crises the market part of MSFT's volatility explodes while the firm-specific part rises much less, so the approximate market share jumps to 0.64 in the financial crisis and 0.87 in the COVID crash. In those months, MSFT's volatility really was mostly the market moving. Outside crises the claim fails: in a calm year like 2017 the share is 0.32, and in 2000–02 it stayed near its average of 0.36 even though MSFT's volatility was very high, because the antitrust case and the dot-com unwinding raised its firm-specific volatility as much as its market volatility. Year-by-year regressions show how unstable the relationship is: the R^2 ranges from 0.14 (2006) to 0.75 (2018) and the beta from 0.73 to 2.30. The R^2 has drifted up over time, to 0.64–0.75 in 2018–2020, as Microsoft became one of the largest weights in the index itself.

Verdict. The claim is wrong as a description of MSFT over 1990–2020. The market explains 41% of MSFT's variance unconditionally, and the approximate conditional share averages 0.36, and the firm-specific remainder is large, persistent and predictable in its own right. The claim is a reasonable approximation only in market-wide crises and, increasingly, in the most recent years.



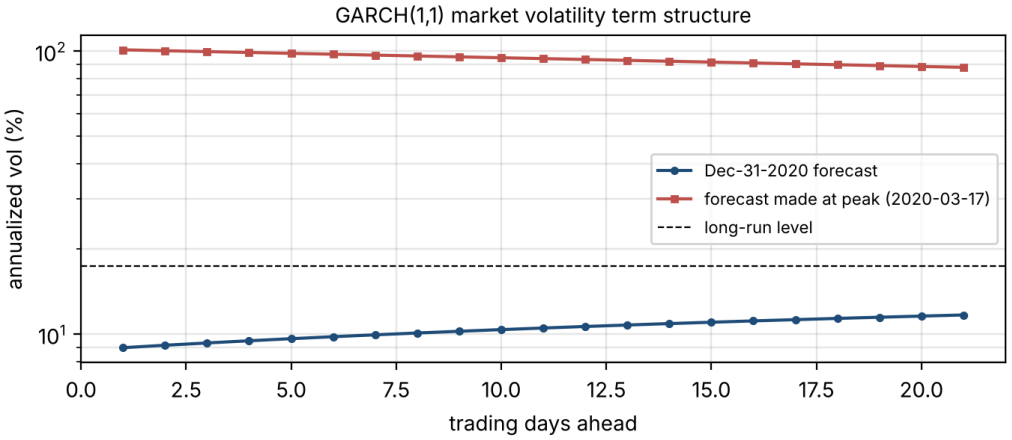
MSFT variance decomposition. Top: market and firm-specific parts of MSFT's volatility. Bottom: approximate market-related share of MSFT's conditional variance (dashed line = sample mean) and yearly market-model R²

(d) What the model says about market risk over the next month (as of 31 Dec 2020)

Under GARCH(1,1), the h-step variance forecast is $E_t\sigma^2_{t+h} = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \bar{\sigma}^2)$. The one-month (21-trading-day) variance is the sum of these 21 forecasts, which gives a one-month standard deviation of 3.0% (5.0% at the long-run level). The ranges and loss probabilities below come from **simulation**, not from a normal approximation. Even with Gaussian daily shocks, the sum of 21 GARCH returns is not exactly Gaussian, because each day's variance depends on the previous day's shock. I simulate 200,000 21-day paths of the fitted model, starting from the 31 Dec 2020 state with Gaussian daily innovations, and read off the quantiles of the cumulative return. The simulated 95% range is −4.5% to +7.5%, the 5% quantile is a loss of 3.4%, and the 1% quantile is a loss of 6.0%. A normal approximation with the same mean and variance gives almost the same numbers (−4.4% to +7.4%) because at this low starting volatility the month's variance changes little. The half-life refers to deviations of the *variance* from its long-run level, not of volatility.

- **How the model thinks.** The model predicts how large tomorrow's market swings will be from two things: how large today's move was, and how turbulent the market has been recently. A big move raises the forecast immediately and a run of quiet days lowers it, but the forecast is always pulled back toward the long-run average of about 17% a year, because neither turbulent nor quiet periods last forever.
- **What it says now.** After a quiet end to 2020, the model puts current market risk at about 9% a year, roughly half the normal level. The pull back toward normal is slow: the gap between today's risk and normal, measured in variance, halves about every two months. So risk is expected to stay low next month, rising only to about 12% by the end of January. In return terms, a typical monthly move is about 3% up or down, against 5% in an average month. In 19 of 20 simulated months starting from today's conditions, the market ends between about −4.5% and +7.5%; a loss worse than 3.4% happens about 1 month in 20, and a loss worse than 6% about 1 month in 100.

- **What could go wrong.** Low forecast risk is not the same as no risk. A single large shock would raise all of these numbers at once: on 17 March 2020 the same model put annualized risk near 100% and a one-month standard deviation of about 27%. The simulation also assumes bell-shaped daily surprises, while real daily returns have more extreme days than that, so these loss probabilities tend to understate the chance of a crash.



Market volatility term structure

Question 2 (PhD) — Gaussian GARCH-in-mean

Model and MLE objective.

$$r_t = \mu + \lambda\, g(\sigma_t^2) + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t,\ z_t \sim N(0,1), \qquad \sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2,$$

where $g(\sigma^2) = \sigma^2$ (the risk–return trade-off) or $g = \sigma$ or $\log \sigma^2$. With $\theta = (\mu, \lambda, \omega, \alpha, \beta)$,

$$\hat{\theta} = \arg\max_{\theta}\ \mathcal{L}(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left[\log 2\pi + \log \sigma_t^2(\theta) + \frac{(r_t - \mu - \lambda g(\sigma_t^2(\theta)))^2}{\sigma_t^2(\theta)}\right]$$

subject to $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta < 1$. The key difference from plain GARCH is that the residual $\varepsilon_t(\theta) = r_t - \mu - \lambda g(\sigma_t^2)$ now depends on the variance parameters, so the two recursions must be run jointly: compute σ_t^2 from ε_{t-1}, then ε_t from σ_t^2. For the same reason the information matrix is no longer block-diagonal between the mean and variance parameters. A misspecified variance equation now makes the mean estimates inconsistent too, so the two parts must be estimated jointly rather than in two steps.

Illustration (not required). With $g = \sigma^2$, the market gives $\hat{\lambda} = 0.041$ (robust s.e. 0.013) and MSFT gives $\hat{\lambda} = 0.012$ (s.e. 0.009). The market shows a significant positive risk–return trade-off and MSFT does not, which is what theory predicts if only systematic risk is priced.

Adding return predictors X_t to the mean. Replace the mean with $m_t(\theta) = \mu + \lambda g(\sigma_t^2) + \gamma' X_{t-1}$, where X_{t-1} is known at $t-1$ (dividend yield, term spread, lagged returns, …), and maximize

$$\mathcal{L}(\theta, \gamma) = -\tfrac{1}{2}\sum_t\left[\log 2\pi + \log \sigma_t^2 + \frac{(r_t - m_t)^2}{\sigma_t^2}\right], \qquad \varepsilon_t = r_t - m_t.$$

Only two things change: the extra parameters γ enter the residual ε_t, and the variance recursion is fed this new residual. The predictors must be known before the return they predict, and they must not be collinear with $g(\sigma_t^2)$. If one of them is itself a volatility proxy, λ and γ compete for the same variation and become weakly identified.

Adding variance predictors Q_t to the GARCH equation (GARCH-X).

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \delta' Q_{t-1}$$

Examples are lagged VIX², realized variance, or market variance as in Q1(c). The objective is unchanged, but the model needs four additional regularity conditions, because the restrictions that made plain GARCH well behaved no longer cover the new term.

The first is positivity. In plain GARCH, $\omega > 0$ and $\alpha, \beta \geq 0$ guarantee $\sigma_t^2 > 0$, but $\delta' Q_{t-1}$ can be negative and push the variance below zero, at which point the likelihood is undefined. Positivity is guaranteed if the predictors are non-negative (variances, squared spreads) and $\delta \geq 0$. For predictors that can take either sign, one must either restrict $\omega + \delta' Q_t > 0$ over the whole support of Q, or reparameterize so positivity holds automatically, for example with $(\delta' Q)^2$, $\exp(\delta' Q)$, or an EGARCH-X model for $\log \sigma_t^2$.

The second is measurability. Q_{t-1} must be known at $t-1$. Otherwise σ_t^2 is not a conditional variance, the prediction-error decomposition behind the likelihood breaks down, and the model is no longer a forecasting model.

The third is stationarity. With an exogenous input, $\alpha + \beta < 1$ is no longer sufficient on its own. The variance inherits the behaviour of Q, so Q_t must itself be strictly stationary and ergodic with $E|Q_t| < \infty$. Then $E\sigma^2 = (\omega + \delta' E[Q])/(1 - \alpha - \beta)$ exists. A trending or unit-root predictor would make σ_t^2 non-stationary even with $\alpha + \beta < 1$. These conditions deliver a well-defined model and, with identification, consistency of the (Q)MLE. They do not by themselves deliver asymptotic normality. That additionally requires the true parameter to lie in the interior of a compact parameter space, finite fourth moments of the innovations ($Ez_t^4 < \infty$, so that the score has a finite variance), and enough moments of Q_t and σ_t^2 for the score and Hessian to satisfy a law of large numbers and a central limit theorem (see Francq and Zakoïan, 2004, for GARCH and Han and Kristensen, 2014, for GARCH-X).

The fourth is identification. ω and $\delta' E[Q]$ can be told apart only if Q actually varies over time, and Q must not be (even asymptotically) collinear with ε_{t-1}^2 or σ_{t-1}^2. A related issue arises in testing. If positivity is imposed through $\delta \geq 0$, the null $\delta = 0$ lies on the boundary of the parameter space. For a scalar δ with all other parameters in the interior, the likelihood-ratio statistic is then asymptotically $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ rather than χ_1^2 (Andrews, 2001). The mixture weights change with the number of restricted parameters and their correlations, and if another parameter is also on its boundary (for example $\alpha = 0$, which is common when Q is a strong volatility proxy) the limit changes again. If δ is not sign-restricted, $\delta = 0$ is an interior point and the usual χ^2 applies.

Question 3 (PhD) — GJR-GARCH(1,1,1) with Student-*t* innovations

Model.

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \overset{iid}{\sim} t_\nu^{std} \text{ (mean 0, variance 1, } \nu > 2),$$
$$\sigma_t^2 = \omega + (\alpha + \gamma\, \mathbb{1}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2.$$

The log-likelihood with the standardized-t density is

$$\mathcal{L} = \sum_t [\log \Gamma(\tfrac{\nu+1}{2}) - \log \Gamma(\tfrac{\nu}{2}) - \tfrac{1}{2}\log(\pi(\nu-2)) - \tfrac{1}{2}\log\sigma_t^2 - \tfrac{\nu+1}{2}\log\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right)].$$

Non-integration (covariance-stationarity) condition. Write $\varepsilon_{t-1}^2 = \sigma_{t-1}^2 z_{t-1}^2$ and $\mathbb{1}\{\varepsilon_{t-1} < 0\} = \mathbb{1}\{z_{t-1} < 0\}$ (since $\sigma_{t-1} > 0$). Then

$$\sigma_t^2 = \omega + \sigma_{t-1}^2 [\alpha z_{t-1}^2 + \gamma z_{t-1}^2 \mathbb{1}\{z_{t-1} < 0\} + \beta].$$

Because z_{t-1} is independent of $\sigma^2_{t-1} \in \mathcal{F}_{t-2}$,

$$E_{t-2}\,\sigma_t^2 = \omega + (\alpha Ez^2 + \gamma E[z^2 \mathbb{1}\{z<0\}] + \beta)\sigma^2_{t-1} = \omega + \phi\,\sigma^2_{t-1}.$$

Here $Ez^2 = 1$. Because the standardized t is symmetric about zero, $E[z^2\mathbb{1}\{z<0\}] = \frac{1}{2}Ez^2 = \frac{1}{2}$, for any $\nu > 2$. Hence

$$\phi = \alpha + \tfrac{\gamma}{2} + \beta.$$

Iterating, $E_{t-k}\sigma_t^2 = \omega\sum_{j=0}^{k-2}\phi^j + \phi^{k-1}\sigma^2_{t-k+1}$. Shocks to variance die out, and the unconditional variance $E\sigma_t^2 = E\varepsilon_t^2 = \omega/(1-\phi)$ is finite and positive, if and only if

$$\alpha + \tfrac{1}{2}\gamma + \beta < 1 \qquad (\text{with } \omega>0,\ \alpha\ge 0,\ \alpha+\gamma\ge 0,\ \beta\ge 0,\ \nu>2).$$

At $\phi = 1$ the model is *integrated* (IGARCH-type): variance shocks are permanent, and the unconditional variance does not exist. Strict stationarity needs only the weaker condition $E\log(\alpha z^2 + \gamma z^2\mathbb{1}\{z<0\} + \beta) < 0$; the condition above is the one for a finite unconditional variance and mean-reverting forecasts. With asymmetric innovations, $\frac{1}{2}$ would be replaced by $E[z^2\mathbb{1}\{z<0\}]$. The forecast half-life is $\log 0.5/\log\phi$.

Estimates (robust s.e. in parentheses; $\phi = \alpha + \gamma/2 + \beta$):

	μ	ω	α	γ	β	ν	ϕ	**log L**
Market	0.065	0.0137	0.009	0.163	0.895	6.9	0.985	−9,832.4
	(0.008)	(0.0024)	(0.006)	(0.018)	(0.010)	(0.57)	(0.003)	
MSFT†	0.073	0.032	0.056	0.068	0.910	5.3	0.99999	−15,220.2
	(0.016)	(0.010)	(0.010)	(0.015)	(0.014)	(0.32)	—	

† Estimation imposes $\phi \le 1 - 10^{-5}$, and for MSFT this constraint binds: $\hat\phi = 0.99999$. Dropping the constraint gives $\hat\phi = 1.0017$ with a log-likelihood gain of only 0.12 (LR = 0.24), so the data cannot distinguish MSFT's GJR-t variance from an integrated, or even mildly explosive, process. Because the estimate is on the boundary of the parameter space, the usual interior asymptotics do not apply. The MSFT standard errors are computed as if the estimate were interior and are indicative only, and no standard error is reported for $\hat\phi$. For the market, $\hat\phi = 0.985$ (s.e. 0.0035) is interior and significantly below one ($t = 4.3$).

Log-likelihood ladder (which feature matters):

	GARCH-N	**GARCH-*t***	**GJR-N**	**GJR-*t***
Market	−10,110.2	−9,927.5	−9,999.7	−9,832.4
MSFT	−15,649.6	−15,238.1	−15,624.0	−15,220.2

GJR-*t* compared with GARCH(1,1)-N. The richer model fits far better. Adding two parameters raises the log likelihood by 278 for the market and 429 for MSFT, which is enormous by any likelihood-ratio standard. The ladder shows where the gain comes from, and the answer differs between the series. For MSFT almost all of it comes from fat tails (+412 from switching to t innovations, against +25 from the leverage term), whereas for the market both features matter (+183 from fat tails, +111 from leverage). The estimated degrees of freedom confirm the fat tails: $\hat\nu = 6.9$ for the market and 5.3 for MSFT. Even after GARCH scaling, the standardized shocks are far from Gaussian, and more so for the single stock, whose firm-specific jumps are exactly what part (a) pointed to. This is also why the Gaussian GARCH of Q1 has to be read as a quasi-MLE.

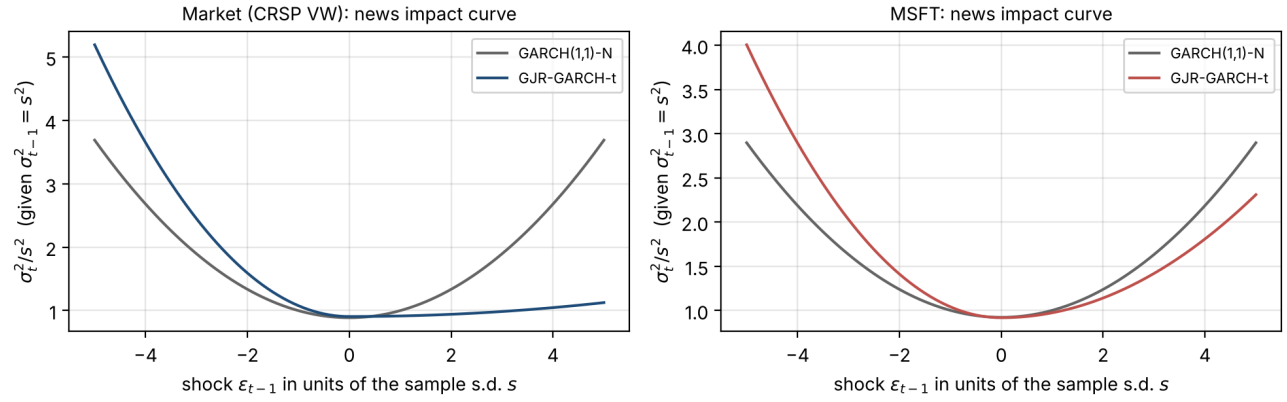
The dynamics change in an instructive way. In GARCH-N, all news enters through one symmetric coefficient α. In GJR, the market's symmetric coefficient collapses to $\alpha = 0.009$, which is not significant, while the negative-news

coefficient is $\gamma = 0.163$. Essentially *all* of the market's ARCH effect therefore comes from negative returns; GARCH-N had been averaging a strong reaction to bad news with almost no reaction to good news. The market's persistence is unchanged at $\phi = 0.985$ (s.e. 0.003; the variance half-life is about 46 days), but its implied long-run volatility falls from 17.4% to 15.2%. The asymmetric model attributes high-variance periods to runs of negative shocks rather than to a higher baseline. For MSFT the persistence rises to the unit-root boundary, because both the average news response $\alpha + \gamma/2$ (0.079 → 0.090) and β (0.906 → 0.910) go up. Under a fat-tailed density, MSFT's largest outliers are down-weighted as ordinary t-draws instead of forcing a lower β, which leaves a smoother and more persistent variance path. As a result, MSFT's long-run variance is not identified by this model: with $\hat{\phi}$ on the boundary, $\omega/(1-\phi)$ is either undefined or extremely unstable.

Leverage: market vs MSFT.

	α **(good news)**	$\alpha + \gamma$ **(bad news)**	**Ratio bad/good**
Market	0.009	0.171	≈ 19
MSFT	0.056	0.124	≈ 2.2

The asymmetry is far stronger for the market. Its $\gamma = 0.163$ is 2.4 times MSFT's 0.068. For the index, good news barely raises volatility at all, while for MSFT both signs matter. This is the standard finding, and it argues against a pure *financial leverage* explanation. Microsoft had essentially no debt over this period, yet its volatility asymmetry is mild, while the index's is extreme. The asymmetry instead looks systematic. Under **volatility feedback**, a rise in market risk raises required returns and lowers prices, so negative returns and rising volatility arrive together; correlations also spike in selloffs. Both effects act on the aggregate. Firm-specific shocks, by contrast, raise MSFT's volatility whether they are good or bad: the October 2000 earnings surprise (+19.6%) was as much a volatility event as the April 2000 antitrust losses. Consistent with Q1(c), MSFT's γ is 0.42 times the market's, close to the approximate 0.36–0.41 market-related share of its variance, so MSFT's leverage effect is about what its market component alone would deliver.



News impact curves: next-day variance after a shock ε_{t-1}, given that yesterday's variance equals the sample variance s^2. Both axes use the sample s.d. s rather than the long-run s.d., which does not exist for MSFT's GJR-t estimate because $\hat{\phi}$ is at the boundary.